

DG Interns Hub – Cybersecurity Internship



WEEK 5 VAPT TOOLS Practical Lab

Author: B Sai Amulya Goud

Intern Id: DG/SEPTEMBER/CYBER/179

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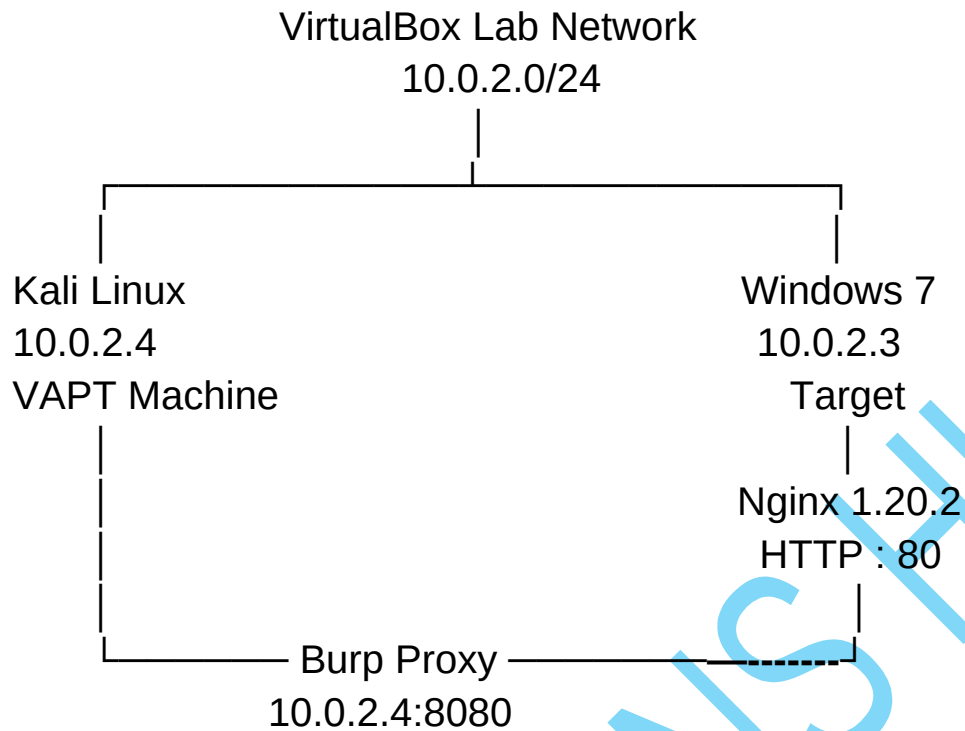
OBJECTIVES

- To understand the basic concepts of **Vulnerability Assessment and Penetration Testing (VAPT)**.
- To learn how to use **Nmap** for network scanning, open-port identification, and service detection.
- To use **Nikto** to identify common web-server security issues and missing security configurations.
- To learn the basics of **Burp Suite** as a web application security testing and HTTP proxy tool.
- To capture and analyze **HTTP requests, headers, and cookies** using Burp Suite.
- To identify and document at least **three security findings** from the practical testing.
- To understand security weaknesses in a controlled and authorized laboratory environment.
- To gain hands-on experience with commonly used **VAPT tools** while following ethical and authorized testing practices.

Lab Setup :

The Kali Linux machine was used as the **security testing machine**, while the Windows 7 virtual machine hosted the Nginx web server as the **authorized target**.

The assignment specified <http://testphp.vulnweb.com> as the safe external target. However, during testing it was unreachable on port 80 from the Kali environment, so the **Nikto and Burp practical demonstrations were performed against the user's own authorized Windows 7/Nginx lab (10.0.2.3)**. This should be stated honestly in the report rather than attributing the local findings to VulnWeb.



Component	Details
Attacker/Testing Machine	Kali Linux
Target Machine	Windows 7
Virtualization	Oracle VirtualBox
Kali IP Address	10.0.2.4
Windows 7 IP Address	10.0.2.3
Network	10.0.2.0/24
Web Server	Nginx 1.20.2
Web Service	HTTP – Port 80
Network Scanner	Nmap 7.98
Web Vulnerability Scanner	Nikto 2.6.0
Web Proxy/Testing Tool	Burp Suite Community Edition
Target Web Page	Nginx default "Welcome to nginx!" page

TASK 1 NMAP NETWORK SCANNING

- `nmap --version` & `nmap -sS testphp.vulnweb.com` :

```
(kali@kali)-[~]
$ nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.5 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
$ nmap -sS testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-02 13:33 -0400
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.00057s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.69 seconds
```

- `nmap -sV testphp.vulnweb.com` :

```
(kali@kali)-[~]
$ nmap -sV testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-02 13:39 -0400
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.00078s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.21 seconds
```

- `nmap -A testphp.vulnweb.com` :

```
(kali@kali)-[~]
$ nmap -A testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-02 13:41 -0400
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0019s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-380U, DP-6310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100 S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-3000P wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.09 ms ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.78 seconds
```

TASK 2 Nikto Web Vulnerability Scan

TARGET : 10.0.2.3

Server: nginx/1.20.2

RESULT : 8218 requests: 0 errors and 7 items reported

```
(kali@kali):~$ nikto -h http://10.0.2.3
- Nikto v2.6.0

+ Target IP:      10.0.2.3
+ Target Hostname: 10.0.2.3
+ Target Port:    80
+ Platform:       Unknown
+ Start Time:     2020-10-03 08:12:31 (GMT-4)

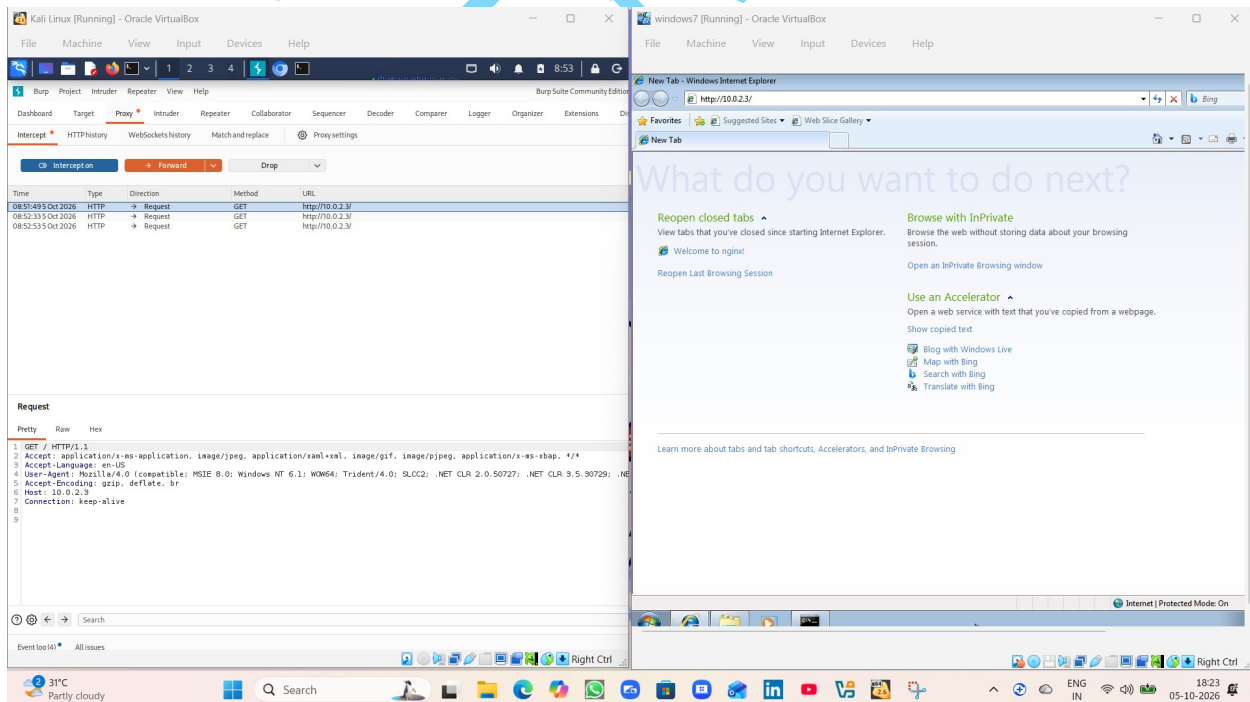
+ Server: nginx/1.20.2
+ ERROR: Failed to check for updates: 403
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ 8218 requests: 0 errors and 7 items reported on the remote host
+ End Time:     2020-10-03 08:12:49 (GMT-4) (18 seconds)

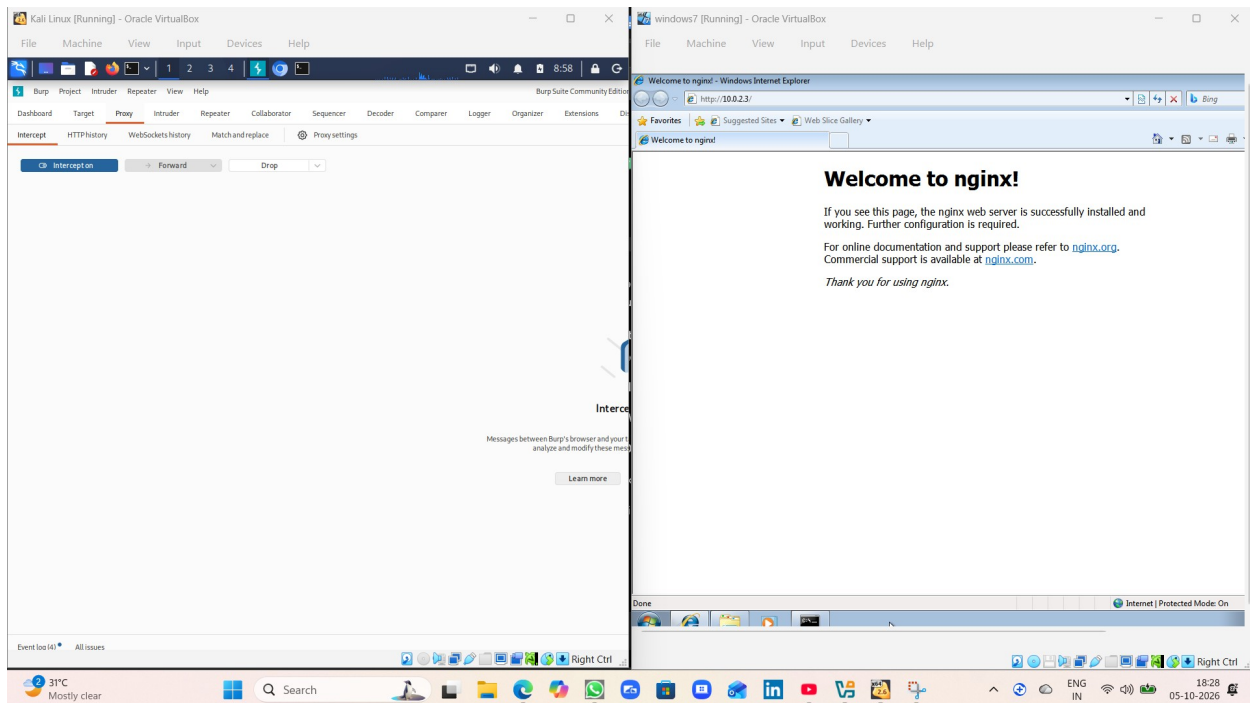
+ 1 host(s) tested
```

Finding	What it means
Missing Permissions-Policy header	A recommended browser security header is not configured.
Missing Referrer-Policy header	The server doesn't specify how much referrer information browsers should send.
Missing Content-Security-Policy (CSP)	A major browser security control is not configured.
Missing X-Content-Type-Options	The server doesn't explicitly prevent MIME-type sniffing.
Missing Strict-Transport-Security (HSTS)	HSTS is not configured.
X-Frame-Options deprecated	Nikto reports this header as deprecated and recommends using CSP frame-ancestors .

TASK 3 BURP SUITE

- Burp Suite was used as an **HTTP proxy and web security testing tool** to intercept and analyze communication between the Windows 7 browser and the Nginx web server.
- A Burp Proxy listener was configured on **10.0.2.4:8080**, and the Windows 7 browser was configured to route HTTP traffic through the Kali machine. HTTP requests to the authorized lab target **10.0.2.3** were successfully intercepted and analyzed.
- The captured requests were examined for details such as the **HTTP method, host, user-agent, headers, and connection information**. Burp Suite's HTTP History was also used to record multiple requests and their responses, including successful HTTP responses from the Nginx server.





• HTTP Request History :

The image shows the Burp Suite HTTP History tab. The selected request is a GET request to http://10.0.2.3, which returned a 200 status code and an HTML response. The request details are shown in the Request pane, and the response is shown in the Inspector pane.

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Time	Listener port	Start response time
1	http://10.0.2.3	GET	/										10.0.2.3	08:24:58.5 Oct 2...	8080	
2	http://10.0.2.3	GET	/										10.0.2.3	08:24:58.5 Oct 2...	8080	
4	http://10.0.2.3	GET	/										10.0.2.3	08:29:24.5 Oct 2...	8080	
6	https://www.google.com	GET	/warmup.html			404	1782	HTML	html	Error 404 (Not Found)!!1		✓	142.251.154.119	08:28:57.5 Oct 2...	8080	113
7	http://10.0.2.3	GET	/										10.0.2.3	08:30:45.5 Oct 2...	8080	
8	http://10.0.2.3	GET	/										10.0.2.3	08:31:02.5 Oct 2...	8080	
9	http://10.0.2.3	GET	/			200	850	HTML		Welcome to nginx!			10.0.2.3	08:31:49.5 Oct 2...	8080	1
10	http://10.0.2.3	GET	/			200	850	HTML		Welcome to nginx!			10.0.2.3	08:32:33.5 Oct 2...	8080	3
11	http://10.0.2.3	GET	/			200	850	HTML		Welcome to nginx!			10.0.2.3	08:33:53.5 Oct 2...	8080	2
12	http://10.0.2.3	GET	/			200	850	HTML		Welcome to nginx!			10.0.2.3	08:37:24.5 Oct 2...	8080	
13	http://10.0.2.3	GET	/			304	180						10.0.2.3	09:01:56.5 Oct 2...	8080	5

Request

```

1 GET / HTTP/1.1
2 Accept: application/x-ms-application, image/jpeg, application/xaml+xml, image/gif, image/png, application/x-ms-xbap, */*
3 Accept-Language: en-US
4 User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729)
5 Accept-Encoding: gzip, deflate, br
6 Host: 10.0.2.3
7 Connection: keep-alive
8
9

```

Inspector

Request attributes: 2

Request headers: 6

CONCLUSION

- I Learned basic Nmap scanning techniques for host and service discovery
- I Identified web-server misconfigurations using Nikto Captured and analyzed HTTP requests using Burp Suite Understood ports, services, and HTTP methods
- I Learned to interpret results rather than only running commands
- Understood the importance of authorization and ethical testing

This practical provided hands-on experience with the basic VAPT workflow using Nmap, Nikto, and Burp Suite. The assessment identified network accessibility information, web-server security misconfigurations, and security-header issues in an authorized local lab.

No destructive exploitation was performed. All testing was conducted strictly within an authorized local lab environment.

THANK YOU